

Housing Boom and Capital Allocation within the Firm: Evidence from China*

Xiaoshu Wang¹, Lianxing Yang^{†2}, Meilan Zhang^{‡3}, and Yao Tang⁴

¹School of Economics, Peking University, Beijing, China.

²School of Economics and Management, East China Normal University, Shanghai, China.

³Stanford Center on China's Economy and Institutions, Stanford University, California, America.

⁴Guanghua School of Management, Peking University, Beijing, China.

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Abstract

This paper examines the impact of China's Shantytown Renovation Programs (SRPs)-driven housing booms on firms' capital allocation. Using prefecture-level SRP data and firm-level panel data, we identify a within-firm capital allocation channel: SRPs, particularly cash-based resettlement, induce housing price surges, prompting firms to reallocate capital from productive assets to real estate, especially those with lower returns on productive investments. Instrumental variable regressions confirm the causal link, showing 1% additional planned SRP units reduce firms' productive investment by 0.27%. Further, such reallocation increases industry-level TFP dispersion and firm underinvestment, causing resource misallocation. This study complements existing literature by highlighting an underexplored mechanism, differing from credit-based channels (collateral or crowding-out), and extends understanding of housing policies' firm-level effects.

JEL classification: D2, G3, R3

Keywords: shanty-town restoration, investment in productive assets, real estate investment

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[†]Co-Corresponding author. School of Economics and Management, East China Normal University, No. 3663 North Zhongshan Rd, Putuo District, Shanghai, China. E-mail: yanglianstar@163.com

[‡]Corresponding author. Stanford Center on China's Economy and Institutions, Stanford University, 450 Jane Stanford Way, Stanford, CA 94305, America. E-mail: mlzjyl@126.com

1 Introduction

During the past three decades, housing booms and busts in the United States, China, the United Kingdom, Spain, Ireland, and other countries have profoundly affected national economies and even the global economy. In the vast academic literature devoted to housing cycles, an important strand examines the effects of housing booms on the behavior of firms. Within this literature, one prominent channel is the collateral effect, which suggests that rising housing prices improve the value of collateralizable assets and the net worth of firms, thereby relaxing financial constraints and increasing access to credit (Gan, 2007; Iacoviello and Neri, 2010; Chaney et al., 2012). Another channel operates through credit supply, whereby banks during housing booms are shown to favor borrowers in the real estate sector and reduce the supply of credit to firms in other sectors (Chakraborty et al., 2018; Martín et al., 2021). These reallocations of credit across sectors and firms can cause efficiency losses in the aggregate economy (Miao and Wang, 2014; Jia et al., 2021; Liu et al., 2026).

In this paper, we argue for the importance of an intuitive yet underexplored channel: within-firm capital allocation. Theoretically, Dong et al. (2022) argue that surging housing prices can induce firm owners to endogenously reallocate capital toward investments in real estate to chase capital gains, thereby reducing investment in productive assets. As a housing boom incentivizes such reallocation on a large scale, aggregate economic efficiency would suffer. However, to the best of our knowledge, there exists scant empirical evidence that housing booms depress firms' investment in productive assets through this reallocation channel.

To empirically analyze the within-firm capital allocation channel, we examine the effects of the housing boom from 2016 to 2020 in China using detailed firm-level data on Chinese firms. The housing boom was driven to a substantial extent by the Shantytown Renovation Programs (SRPs), an urban development program that was promoted to national significance in 2015. We show that prefecture-level SRPs, as our proxy for local housing booms, significantly depress firms' productive asset investment rather than stimulating it. We establish causality from housing booms to capital allocation using instrumental variable regressions that exploit the exogenous variation in SRPs associated with the area of urban shantytowns land and China's net exports. The former variable is related to the potential supply of public land used in SRPs and the latter reflects the policymakers' preference for SRPs as a substitute for external

demand. Based on our 2SLS estimates, the elasticity of SRP intensity to local firm's productive asset investment is -0.27.

We show that the decline in firms' investment in productive assets is unlikely to be explained by collateral constraints or a reduction in the supply of bank credit. After introducing interaction terms between SRPs and measures of firms' asset pledgability to account for the potential collateral effects of housing assets, we find that SRPs are still generally associated with a reduction in productive asset investment. Moreover, the negative effect of SRPs on investment remains when we control for the supply of credit or the general availability of funding. After accounting for both collateral and credit crowding-out channel, the evidence supports the within-firm capital allocation channel as an independent mechanism through which housing booms affect firm investment.

Using data on listed firms in China that provide information on housing assets, we then find that firms located in prefectures with more intensive SRPs invest more in real estate. In particular, firms with lower returns on productive assets are more inclined to invest in real estate, consistent with the reallocation conjecture that firms actively weigh between the investment options of productive assets and real estate. Taken together, these findings indicate that housing booms direct firms' internal capital allocation toward real estate at the expense of investment in productive assets.

Finally, we demonstrate that reallocation to more real estate assets results in loss of production efficiency. Following the approach of Hsieh and Klenow (2009), we find that SRP-induced housing booms significantly increases prefecture-industry productivity dispersion which implies increased resource misallocation within industries. Specifically, our estimates indicate that an annual increase of 1% SRP housing units planned leads to a 0.024 rise in prefecture-level industry-level TFP dispersion, a magnitude that amounts to 4.82% of its mean and is thus economically significant. Furthermore, there is evidence of misallocation at the firm level, as housing booms cause local firms to underinvest in productive assets relative to their efficient capital levels.

This paper relates to three strands of the literature. First, our study contributes to the literature on the mechanisms through which housing booms affect firms' investment decisions. Existing studies emphasize either the collateral channel, through which rising housing prices

relax financial constraints and stimulate investment (Gan, 2007; Iacoviello and Neri, 2010; Chaney et al., 2012), or the credit crowding-out effects, whereby housing booms attract bank credit away from the non-housing sector (Chakraborty et al., 2018; Hau and Ouyang, 2018; Martín et al., 2021; Liu et al., 2026). In contrast to these credit-based mechanisms, we highlight a within-firm capital allocation channel, whereby housing booms reshape firm owners’ capital allocation choices between productive assets and real estate.

Several theoretical studies have examined the housing market and firms’ investment in the context of China (Miao and Wang, 2014; Chen and Wen, 2017; Dong et al., 2022). In particular, we are closely related to Dong et al. (2022) who develop a model of firm-level capital allocation choices and show that when negative productivity shocks reduce returns to productive assets, firms will reallocate capital from productive assets toward real estate. While our study centers on a housing price boom rather than a negative productivity shock as the driving force, our firm-level micro evidence broadly supports their theoretical predictions of return-driven capital reallocation toward real estate. Based on data of listed firms in China, Chen et al. (2017) find evidence that rising land values induce firms to increase investment in both productive assets (through the collateral channel) and speculative real estate. Relative to Chen et al. (2017), we use a much larger and more representative set of Chinese firms to show that housing booms, in general, depress investment in productive assets. Thus, the negative effects of housing booms on investment in productive assets may be underestimated in the previous literature.

Second, our paper relates to the literature on firms’ misallocation of resources, notably Hsieh and Klenow (2009) and papers that it motivates. Among studies examining the role of housing booms in generating misallocation, the existing literature has largely focused on sectoral misallocation (Miao and Wang, 2014; Chen et al., 2017; Jia et al., 2021). Recent studies further document between-firm misallocation driven by credit allocation and differential collateral effects (Liu et al., 2026; Basco et al., 2025). In this paper, we highlight the within-firm capital allocation channel as a previously underexplored mechanism through which housing booms generate resource misallocation.

Finally, our paper contributes to the literature on the economic impacts of slum upgrading programs and urban revitalization policies to which China’s Shantytown Renovation Programs (SRPs) can be compared. Related initiatives include the low-cost housing programs (Cam,

1939) and the Low-Income Housing Tax Credit (LIHTC) program (Diamond and McQuade, 2019) in the United States, the housing voucher schemes in India (Barnhardt et al., 2017), and the slum upgrading projects in Latin America (Galiani et al., 2017) and China (He et al., 2024). Existing studies primarily focus on evaluating the effectiveness and cost-benefit outcomes of such programs, particularly their impacts on household-level indicators such as employment, income, and welfare. We extend this literature by shifting the focus from household-level outcomes to firm-level responses, showing that a cash-based housing program generates negative economic effects through depressing firms’ investment in productive assets.

The remainder of the paper is organized as follows. Section 2 provides background information on the institutional setting. Section 3 describes the data. Section 4 presents the baseline estimation results. Section 5 investigates the effects of housing prices on capital misallocation. Section 6 concludes.

2 Institutional Background and Working Hypothesis

2.1 China’s Shantytown Renovation Projects (SRPs)

China’s Shantytown Renovation Projects (SRPs) are large-scale urban redevelopment initiatives aimed at improving housing conditions in deteriorated urban neighborhoods, which are predominantly occupied by low-income households. The shantytown dwellings examined in this paper consist of substandard, makeshift housing units built by private individuals on public land, most commonly land owned by state-owned enterprises (SOEs). The program constitutes a central component of China’s urbanization and social welfare agenda and has been implemented nationwide under unified central policy guidance, with execution delegated to local governments.

In 2013, the State Council issued the *Opinions on Accelerating the Renovation of Shantytowns*, which clearly proposed the renovation of 10 million housing units in various types of shantytowns during 2013–2017. The document also emphasized integrating shantytown redevelopment with the improvement of people’s livelihoods, the expansion of effective investment, and the advancement of a people-centered new urbanization strategy. This indicates that, since 2013, shantytown redevelopment has no longer been merely a localized housing improvement project but has instead been elevated to an important public policy coordinated and promoted

at the central-government level.

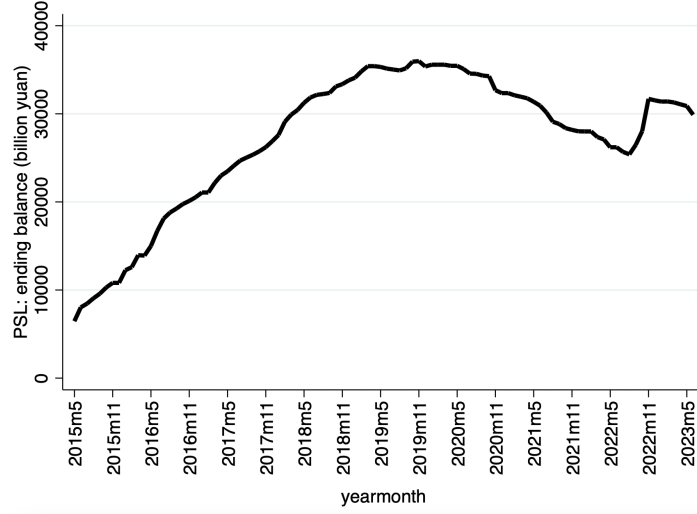
A key institutional component of SRPs is the resettlement mechanism. Two main approaches exist: in-kind resettlement and monetary resettlement. In-kind resettlement provides newly constructed or government-purchased housing for relocated residents, whereas monetary resettlement offers cash compensation that allows households to purchase housing in the market. Before 2015, in-kind resettlement dominated SRPs due to limited housing market development and local governments' preference for centralized relocation.

2.2 Cash-based resettlement and Chinese housing booms

After 2015, monetary resettlement rapidly rose in prominence within shantytown redevelopment, and this shift was closely related to contemporaneous policies aimed at reducing housing inventories in the real estate market. In 2015, the State Council issued the *Opinions on Further Improving Work Related to the Renovation of Urban Shantytowns, Dilapidated Urban and Rural Housing, and Supporting Infrastructure Construction*, which proposed a three-year plan for 2015–2017 under which another 18 million units in various types of shantytowns would be redeveloped. During this stage, shantytown redevelopment not only continued to bear the responsibility of improving people's livelihoods but was also assigned a more explicit role in stabilizing economic growth and reducing the housing inventory in the housing market.

The immediate background to the accelerated promotion of monetary resettlement was the mounting inventory pressure in the commercial housing markets of third- and fourth-tier cities, accompanied by a slowdown in housing sales. In 2015, a marked contrast emerged between housing sales and inventory levels. This situation of persistently high inventories prompted policymakers to begin linking the resettlement demand generated by shantytown redevelopment with the policy objective of reducing housing inventory in the real estate sector. After June 2015, the State Council explicitly promoted monetary resettlement by requiring that the share of monetary resettlement should be no less than 50 percent. The share of monetary resettlement in shantytown redevelopment rose continuously, increasing from 9 percent in 2014 to 54 percent in 2017(He et al., 2024). Figure 1 illustrates the rapid expansion of PSL over this period, highlighting the scale and persistence of policy-driven liquidity injections into the housing sector. This change shows that, within a very short period, monetary resettlement evolved from a marginal approach into one of the dominant modes of shantytown redevelopment.

Figure 1: Ending Balance of PSL in China

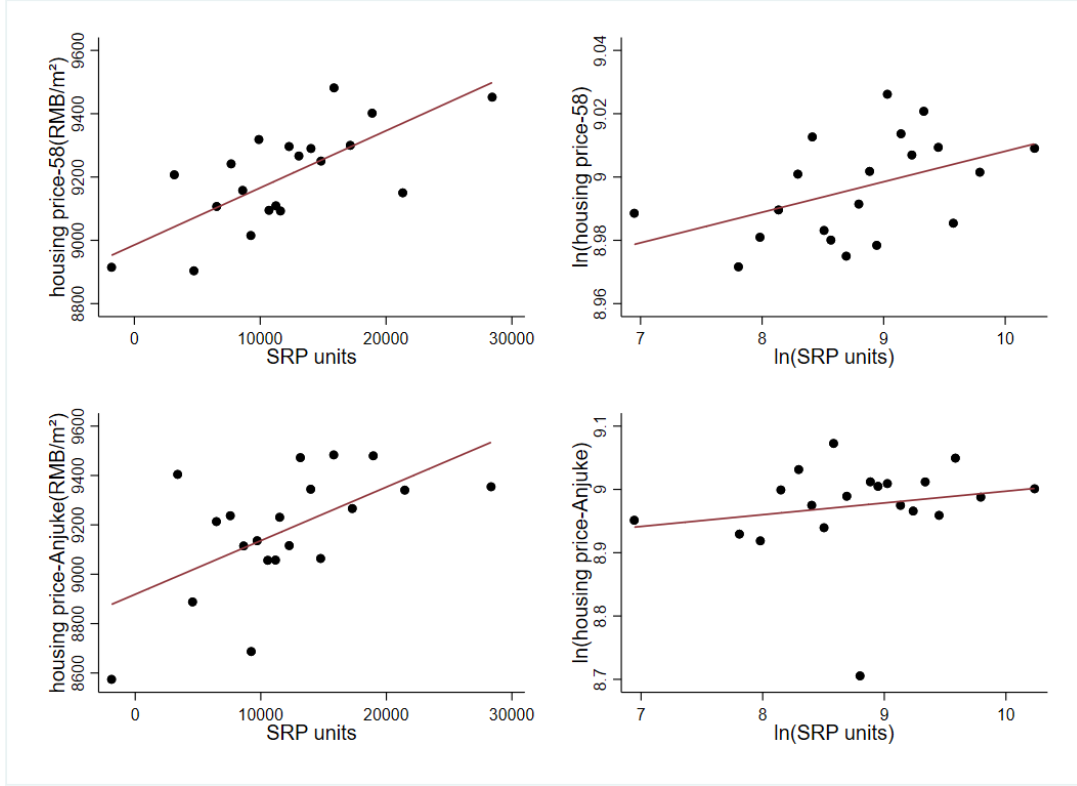


Notes : Data from the People's Bank of China.

Compared with in-kind resettlement, monetary resettlement shortened the resettlement process and generated demand for market-provided housing units. As a result, cities with more intensive SRP implementation experienced larger increases in housing prices. Figure 2 illustrates the positive relationship between SRPs and local housing price growth, highlighting the role of cash-based resettlement as a policy-induced housing demand shock.

However, as housing prices rose rapidly in some regions, policy orientation gradually shifted. In 2017, authorities proposed limiting monetary resettlement in cities where housing inventories had already declined substantially. A joint policy issued in 2018 required cities with housing inventory less than 15 months of sale to reduce the use of cash-based resettlement and rely more on in-kind solutions. From 2019 onward, fiscal support for SRPs was reduced, and the PSL channel was gradually scaled back.

Figure 2: SRPs and Local Housing Price



Notes : The scatter plot is generated using the binscatter method, controlling for city and year fixed effects. SRPs data are collected from the official websites of local Housing and Urban-Rural Development Bureaus in China. Housing price data are obtained from two major Chinese second-hand housing platforms, 58.com (<https://bj.58.com/>) and Anjuke (<https://www.anjuke.com/>). The two panels on the left present scatter plots of the two housing price indicators against the SRPs units, while the two panels on the right present scatter plots the logarithm of the two housing price indicators against the logarithm of SRPs units.

2.3 Housing Booms and Housing Speculation in China

Since the late 1990s, China's real estate market has experienced significant booms, characterized by rapid price increases and intensified speculative activities. Between 1999 and 2021, the average annual growth rate of housing prices was approximately 7.5%, with a peak year-on-year increase of 25.0% in February 2010, and house prices reached their highest level in 2021.

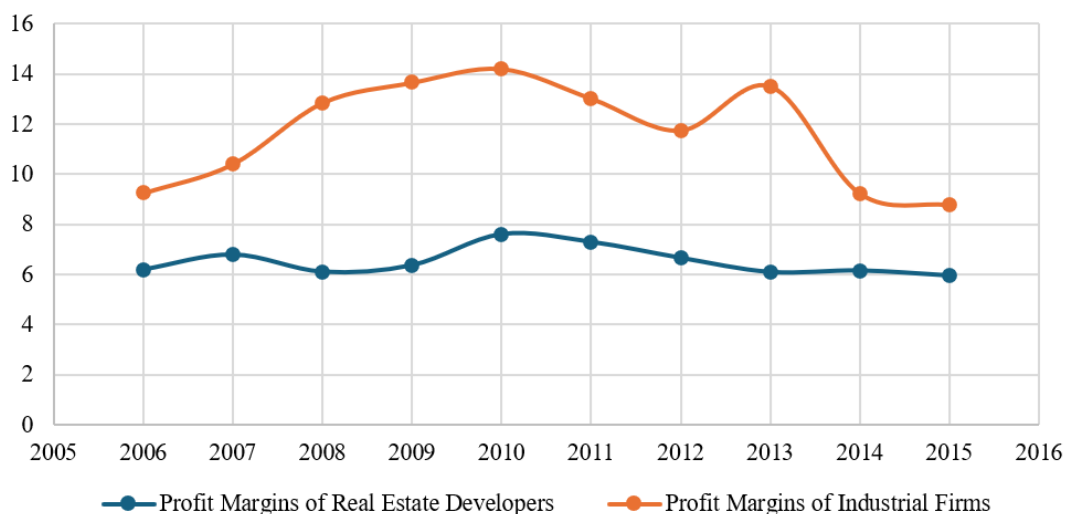
The profitability of the real estate sector has historically outpaced that of the industrial sector. From 2006 to 2015, profit margins in real estate development were consistently higher than those in the industrial sector, incentivizing capital reallocation from industrial enterprises to real estate investments (Figure 3). This divergence in returns not only attracted household investors but also induced non-real-estate firms to engage in housing-related speculative activ-

ities. Empirical evidence indicates that approximately 45% of non-real estate firms in China hold investment properties, which constitute about 15% of their total physical assets (Chen and Wen, 2017). This substantial allocation suggests a strategic shift among firms toward real estate investments, often at the expense of their primary business activities.

The housing booms induced by SRPs further amplified firms' incentives in acquiring real estate. Cash-based resettlement increased housing demand not only from displaced households with genuine housing needs, but also from speculative buyers (firms included) seeking to capitalize on short-term price appreciation. As housing prices rose, real estate increasingly dominated firms' investment calculus, reinforcing the flow of capital from core production activities to real estate investments.

Taken together, we have the following **working hypothesis**: large-scale housing policies, when combined with accommodative financing mechanisms, generate powerful demand-side housing booms that lead firms to allocate more capital to real estate investment at the expense of investment in productive assets. The setting of SRPs provides a natural laboratory to examine how policy-induced housing price shocks affect corporate investment, credit demand, and productivity, which is the central focus of this paper.

Figure 3: Profit Margins of Real Estate Developers and Industrial Firms (%)



Notes : Data from the National Bureau of Statistics of China.

3 Data

3.1 SRPs Data at the City Level

We manually collected annual data on SRPs from the official websites of local Housing and Urban-rural Development Bureaus in China. Our sample covers 97 prefecture-level cities in ten provinces from 2016 to 2022.¹ Since housing prices are forward-looking and our interest lies in the short-run demand-side effects of SRPs, we measure SRP intensity by the annual number of newly planned renovation housing units that is announced to the general public through the official website, rather than completed units.

Between 2016 and 2022, participating provinces planned on average about 110,000 renovation units per year, corresponding to roughly 11,000 units annually at the prefecture level, with some cities exceeding 60,000 units in peak years. Valued at an average market price of nearly one million RMB per unit during 2016–2020,² SRPs imply an annual injection of approximately 10 billion RMB into the housing market of a typical prefecture-level city.

3.2 Firm-level Data of the National Tax Survey Database (NTSD)

Our firm-level data are drawn from the National Tax Survey Database (NTSD), jointly compiled by the Ministry of Finance of the People’s Republic of China and the State Administration of Taxation. The NTSD is a large-scale administrative panel that provides detailed information on firms’ operations, financial statements and tax payments. The data are collected through a centralized reporting system and subject to electronic validation procedures, ensuring a high degree of accuracy and internal consistency.

Compared with commonly used Chinese firm-level datasets, such as listed company databases and the Annual Survey of Industrial Firms, the NTSD offers substantially broader coverage. In addition to publicly listed firms and other large firms, which account for more than 70% of local value-added tax (VAT) revenue, the NTSD also includes a randomly selected sample of smaller firms from the administrative tax registry, thereby capturing a large number of small and medium-sized enterprises (SMEs) (Chen et al., 2023). This feature mitigates the size

¹The ten provinces included in our analysis are Anhui, Shandong, Guangdong, Jiangsu, Henan, Fujian, Shaanxi, Gansu, Guizhou, and Shanxi. Data for the remaining provinces are unavailable or incomplete and are therefore excluded from the analysis.

²The average housing price is calculated based on national residential housing price data from 2016 to 2020, obtained from China Real Estate Information (CREI), <https://www.crei.cn/>.

and listing biases inherent in conventional datasets and enhances the representativeness of the sample.

We merge the NTSD with the SRP dataset and obtain a panel of 50,277 firms from 2016 to 2020 after excluding observations with missing key variables. Descriptive statistics are reported in Table 1.

Table 1: Descriptive Statistics

<i>Sample:</i> 135,807 observations; 50,277 firms						
Variable	Obs.	Mean	Median	SD	Min	Max
Prefecture-level data						
SRPs	212	11783.7311	7470.5000	11511.8474	75	59228
House price_58	210	9198.8333	7522.5000	7689.7105	4254	56934
House price_anju	212	9174.6840	7538	7800.7140	219	57691
Industry-level data						
TFP_Dispersion(75-25)	1959	-1.1098	-1.0166	0.6864	-5.5976	2.0680
TFP_Dispersion(90-10)	1959	-0.4977	-0.3471	0.7826	-5.5976	2.0680
TFP_Dispersion(Std.)	1959	-1.2734	-1.1694	0.7034	-5.9441	1.7214
Firm-level data						
Investment	135807	-0.0201	-0.0533	0.7590	-13.1139	13.3796
Credit	20094	10.1739	10.0345	2.5505	0	20.7449
Iv	135807	12713.2752	11898.1709	5144.5776	1722.8700	24778.0156
RE Investment	1152	0.0334	0.0100	0.0732	0	0.7960
Underinvest1	135807	-0.7070	0.5095	327.6285	-57694.5703	28800.5332
Underinvest2	135807	-0.6700	0.3405	321.5713	-51286.5742	43204.3242
Underinvest3	135807	-0.5764	0.4148	320.8706	-51286.4492	43204.2930
Underinvest4	109287	-0.0004	0.0004	0.6993	-9.5441	12.0032
Ln(value add)	62967	8.1976	8.2636	2.7192	0	17.8366
Ln(TFP)	72225	1.4519	1.5070	0.3967	-6.5418	2.6655
TFP_growth	52746	-0.0167	-0.0050	0.3186	-8.3144	6.4831

Notes: (1) Prefecture-level data report summary statistics for prefecture-level cities over 2016–2022 for which data on SRPs and housing prices are available. SRPs are measured as the annual number of newly planned renovation housing units, collected from the official websites of local Housing and Urban-Rural Development Bureaus for each city. *House price 58* and *House price Anjuke* refer to secondary-market housing prices obtained from 58.com (<https://bj.58.com/>) and Anjuke (<https://www.anjuke.com/>), respectively. (2) Industry-level data report summary statistics for within-industry capital misallocation over 2016–2020. Following Hsieh and Klenow (2009), *TFP Dispersion (75–25)*, *TFP Dispersion (90–10)*, and *TFP Dispersion (Std.)* denote the interquartile range, decile range, and standard deviation of log TFP within an industry, respectively. (3) Firm-level data report summary statistics for firms over 2016–2020 located in prefectures for which SRP data are available. Except for *RE Investment* and *IV*, firm-level variables are constructed from the NTSD database. Specifically, (i) *Investment* is measured as the difference in the logarithm of fixed assets; (ii) *Credit* is measured as the difference in the logarithm of interest-bearing liabilities; (iii) *Underinvest1–Underinvest4* capture firm-level underinvestment following Richardson (2006) and Gan (2007); (iv) *Ln(value add)* is the logarithm of firm-level value added; (v) *Ln(TFP)* and *TFP_growth* denote firm-level TFP estimated following Olley and Pakes (1996) and its logarithmic growth rate. Finally, *RE Investment* is constructed using the Chinese listed firms database and measured as the ratio of investment property to total assets. *Iv* denotes the instrumental variable, defined as the interaction between the area of urban shantytown land in each region as of 2014 and world GDP growth net of China’s GDP growth.

4 Empirical Results

4.1 SRPs and Local Housing Booms

We first establish the relationship between SRPs and local housing booms using prices in the secondary housing market, which better reflect market conditions given China’s strong price regulation in the primary housing market. Following (He et al., 2024), we collect monthly transaction data from 58.com (<https://bj.58.com/>) and Anjuke (<https://www.anjuke.com/>), and aggregate the data to the prefecture level.

Table 2 reports estimates from regressions with city and year fixed effects, showing that SRPs are positively associated with secondary-market housing prices. Quantitatively, an additional 1,000 newly planned renovation units corresponds to about 15–20 RMB per square meter higher local housing prices. Scaling this estimate to the observed variation in SRP intensity, a one-standard-deviation increase in SRP intensity (11,512 units) implies a price increase of 173–231 RMB per square meter, accounting for 50–66% of the average annual growth in secondary-market housing prices. The regression results align with our discussion in section 2, supporting the view that SRPs, particularly those implemented through cash-based resettlement, generate a positive demand shock and a price increase in the local housing market.

4.2 Effect of Housing Boom on Firm Investment in Productive Assets

4.2.1 Empirical Specification

To examine the impact of local housing booms on firm investment, we estimate the following baseline empirical specifications:

$$Investment_{ict} = \beta_1 + \beta_2 SRPs_{ct} + \theta_i + \phi_c + \mu_t + \varepsilon_{ict}$$

where $Investment_{ict}$ is the investment of firm i located in prefecture c in year t . Following Basco et al. (2025), the investment is measured as the difference in the logarithm of fixed assets between year t and $t - 1$. $SRPs_{ct}$ is the logarithm of the announced annual number of newly planned renovation housing units in prefecture c in year t . We include firm, city, and year fixed effects, and cluster standard errors at the industry level.

A key identification concern is that SRP intensity may be endogenous to local economic conditions. For example, prefectures with stronger fiscal capacity and greater political bargaining power vis-à-vis the central government are more likely to obtain SRP funding and support.

Table 2: Housing Price and SRPs

	(1)	(2)	(3)	(4)
	House price_58	House price_58	House price_anju	House price_anju
SRPs	18.0422* (9.9646)	16.6232* (9.5767)	21.7172** (10.7051)	20.1459* (10.3404)
population		14.8351*** (5.4386)		17.9910** (8.7070)
GDP		-0.0514 (0.1444)		-0.1055 (0.1879)
land_area		-1.1969*** (0.4452)		-1.4609** (0.5781)
City FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Dep. Var. Mean	9198.8333	9198.8333	9174.6840	9174.6840
R ²	0.9960	0.9966	0.9934	0.9940
N	210	210	212	212

Notes: (1) Standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. (2) The dependent variable housing prices are obtained from transaction data from 58.com (*house_five*) and Anjuke (*house_anju*), measured in RMB per square meter. The average annual growth rate of housing prices is 349.72 RMB per square meter based on data from 58.com and 350.96 RMB per square meter based on data from Anjuke. (3) The independent variable SRPs is measured as the annual number of newly planned renovation housing units in each prefecture, expressed in thousands. (4) To maintain consistency with the baseline specification, we also estimate regressions using the logarithm of SRPs and the logarithm of housing prices. Both statistical and economic significance remain similar to those reported here. For ease of interpretation, we present results from specifications that regress housing prices directly on the number of renovation units.

These same characteristics are also associated with higher levels of firm investment. As a result, OLS estimates are likely to overestimate the effect of SRPs on investment. To address potential endogeneity, we instrument $SRPs_{ct}$ with the interaction between (i) the area of urban shantytowns land in each region as of 2014, and (ii) world GDP growth net of China's GDP growth. The first component captures cross-regional variation in the initial supply of land eligible for renovation, while the second reflects time-series variation in the central government's incentive to stimulate domestic demand through SRPs in response to external macroeconomic conditions.

The instrument plausibly satisfies the exclusion restriction. The spatial distribution of shantytown land is largely shaped by the historical planning of SOE locations and path-dependent urbanization, while relative world GDP growth is driven by global business cycles and is exogenous to individual firms. Their interaction, therefore, provides a plausibly exogenous source of variation in SRPs.

4.2.2 Baseline 2SLS Results

The OLS and 2SLS regression results of Equation (1) are reported in Table 3. Our regression results indicate that firms located in prefectures with more intensive SRPs implementation exhibit significantly lower levels of productive asset investment. Specifically, a 1% increase in the SRP intensity, which corresponds to roughly 100 additional units in a city in a single year, results in an average decline in local firm’s productive asset investment of approximately 0.01%. Columns (2)–(3) report the 2SLS estimates. The elasticity of SRP intensity to local firm’s productive asset investment is -0.27 and is larger than the OLS estimate, consistent with the expected sign of the bias created by the endogeneity of house prices. Standard diagnostic tests support the relevance of the instrument, and the IV results reinforce the finding that local housing booms induced by SRPs lead to lower firm investment in productive assets.

Table 3: SRPs and Firm Investment in Productive Assets

	FE	first stage	2SLS
	(1)	(2)	(3)
	Investment	Ln(SRPs)	Investment
Ln(SRPs)	-0.0136*** (0.0045)		-0.2735* (0.1421)
IV		0.0001*** (0.0000)	
City FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Dep. Var. Mean	-0.0201	2.1919	-0.0201
CD Wald F		191.9709	
LM_p			0.0002
R ²	0.3967		-0.0391
N	135,807	135,807	135,807

Notes: (1) Standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. (2) The dependent variable is firm investment, measured as the difference in the logarithm of fixed assets between year t and $t - 1$ following Basco et al. (2025). The independent variable is SRPs scale, measured as the logarithm of the announced annual number of newly planned renovation housing units in prefecture c in year t . (3) Column (1) reports OLS estimates, while Columns (2) to (3) report 2SLS estimates. *CD Wald F* denotes the Cragg–Donald Wald F -statistic for the weak identification test. *LM_p* denotes the p-value of the Kleibergen–Paap rk LM statistic for the underidentification test.

4.3 Accounting for Credit-Based Channels

To establish that SRPs affect firm investment through the within-firm capital allocation channel, we need to account for two credit-based effects of SPRs—the collateral effect and the credit crowding-out effect.

Standard collateral-based theories predict that rising land values should expand firms' borrowing capacity and stimulate investment through increased access to credit (Gan, 2007; Chaney et al., 2012). To test this implication, Columns (1)-(3) of Table 4 replace investment with credit as the dependent variable, which is measured as interest-bearing liabilities reported in the NTSD.³ The results show that SRPs significantly reduce firms' credit, i.e., firms located in prefectures with more intensive SRP implementation do not report more borrowing on their books. Quantitatively, an additional 100 newly planned renovation housing units in a city lead to an average reduction of approximately 0.06% in local firms' interest-bearing liabilities. Therefore, even though the housing boom relaxes the collateral constraint, credit obtained by firms does not increase, which indicates a lack of a collateral-pushed credit effect on average.

To further assess whether the effects of collateral on investment in productive assets depend on the level of pledgeable assets, we augment the baseline specification by interacting SRP intensity with firms' ex-ante asset pledgability. Following Basco et al. (2025), we proxy asset pledgability using firms' asset tangibility, measured as the ratio of tangible fixed assets to total assets, and construct an indicator equal to one if a firm's tangibility exceeds the sample median and zero otherwise. Column (4) of Table 4 shows that the interaction term is positive, indicating that the negative effect of SRPs on investment in productive assets is weaker for firms with more pledgeable assets, consistent with the presence of collateral effects. However, the coefficient on SRP intensity remains negative and statistically significant, suggesting that besides the collateral effects, the housing boom has an independent negative effect on investment in productive assets, potentially caused by within-firm capital reallocation in favor of real estate investment.

The other credit-based effect channel is the credit crowding-out channel emphasized by Chakraborty et al. (2018) and Martín et al. (2021). Under this mechanism, housing booms induce banks to reallocate credit toward real estate-related activities, crowding out non-housing firms and dampening their investment. To account for this possibility, we augment the baseline regressions by with regional credit supply conditions, capturing both the aggregate general availability of funding and its allocation toward the real estate sector. Specifically, we include the logarithm of total social financing in each region and control for commercial housing sales,

³According to China's accounting standards, this item includes all liabilities on which firms are required to pay interest, including bank loans and corporate bonds.

Table 4: Accounting for Collateral Effects

	FE	first stage	2SLS	FE
	(1)	(2)	(3)	(4)
	Credit	Ln(SRPs)	Credit	Credit
Ln(SRPs)	-0.0579*** (0.0186)		-0.3547* (0.2000)	-0.0584*** (0.0203)
iv		-0.0005*** (0.0001)		
Ln(SRPs)*collateral_dummy				0.0015 (0.0186)
City FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Dep. Var. Mean	10.1739	1.9948	10.1739	0.0157
CD Wald F		198.8801		
LM_p			0.0000	
R ²	0.8925		-0.0267	0.8925
N	20,094	20,094	20,094	20,094

Notes: (1) Standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. (2) The dependent variable is firm credit, measured as the difference in the logarithm of interest-bearing liabilities between year t and $t - 1$. The independent variable is SRPs scale, measured as the logarithm of the announced annual number of newly planned renovation housing units in prefecture c in year t . (3) Column (1) reports OLS estimates, while Columns (2) and (3) report 2SLS estimates. Column (4) further controls for firms' asset pledgability, measured as the ratio of tangible fixed assets to total assets. We then construct a dummy variable (*collateral_dummy*) that equals one if a firm's tangibility exceeds the sample median, and zero otherwise. This dummy is interacted with the dependent variable and included in the baseline regression.

newly started land development, and government real estate investment, which measure sectoral credit flows directed toward housing-related activities.

Table 5 reports the results with these additional controls. The coefficients on SRP intensity remain statistically significant, although their magnitudes are modestly attenuated relative to the baseline estimates. After controlling for both the aggregate availability of credit and its allocation toward the real estate sector, the negative effect of SRPs on local firms' productive asset investment becomes smaller compared to the baseline results. This evidence indicates that the effects of SRPs are not driven by fluctuations in overall credit supply or by a reallocation of credit toward the real estate sector, suggesting the presence of independent transmission mechanisms through which SRPs influence both firm investment and access to credit.

4.4 Effect of Housing Boom on Firm Investment in Real Estate

In this subsection, we present evidence that rising real estate prices induce firms to reallocate capital toward housing-related investments, resulting in underinvestment in productive assets.

A broad theoretical literature provides foundations for this mechanism. In environments

Table 5: Accounting for Credit Supply Effect

	(1)	(2)	(3)
	Investment	Investment	Investment
Ln(SRPs)	-0.0121*** (0.0045)	-0.0101** (0.0049)	-0.0100** (0.0048)
Ln(social financing)	0.2056** (0.1034)		0.1844* (0.1069)
Ln(commercial housing sales)		-0.0517** (0.0242)	-0.0473* (0.0245)
Ln(new land development)		0.1931 (0.2021)	0.0686 (0.2095)
Ln(government RE investment)		0.0613* (0.0326)	0.0577* (0.0328)
City FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Dep. Var. Mean	-0.0201	-0.0201	-0.0201
R ²	0.3967	0.3968	0.3968
N	135,807	135,807	135,807

Notes: (1) Standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. (2) The dependent variable is firm investment, measured as the difference in the logarithm of fixed assets between year t and $t - 1$. The independent variable is SRPs scale, measured as the logarithm of the announced annual number of newly planned renovation housing units in prefecture c in year t . (3) $Ln(\text{social financing})$ refers to the logarithm of total social financing in each province, compiled by the People's Bank of China. $Ln(\text{commercial housing sales})$, $Ln(\text{new land development})$, and $Ln(\text{government RE investment})$ denote the logarithms of commercial housing sales, newly started land development, and government real estate investment in each prefecture, respectively, compiled by the local statistical bureaus.

with borrowing constraints and incomplete financial markets, firms may use price-inflated assets as vehicles for intertemporal savings and investment (Tirole, 1985). In the context of China, Chen and Wen (2017) show that borrowing-constrained firm owners face a capital allocation choice between productive capital and housing, and rising housing returns tilt capital allocation toward real estate. Similarly, Dong et al. (2022) demonstrate that housing price fluctuations alter the relative returns to housing and productive capital, shaping firms' capital allocation decisions. Behavioral theories further suggest that firms may extrapolate past housing price growth and form overly optimistic expectations about future returns, reinforcing speculative investment in real estate (Barberis et al., 2015; Greenwood and Shleifer, 2014).

Because the NTSD dataset does not provide information on firms' investment in real estate, we rely on the database of listed firms to provide direct evidence that housing booms induce firms to increase their real estate investment. We use investment property as reported in Chinese listed firms' annual reports as the dependent variable, which captures housing properties held by firms to earn rentals or for capital appreciation. Table 6 reports the results. To account for firms' incentives to invest in productive assets, we control for firms' returns to productive

assets, proxied by asset turnover, Tobin's Q, and indicators for persistent losses. Specifically, we construct indicator variables for asset turnover and Tobin's Q that equal one if the firm's value exceeds the sample median and zero otherwise. We also define a loss indicator that equals one if a firm reports negative net profits for two consecutive years, and zero otherwise.

In all three specifications, the estimated effect of housing booms on firms' real estate investment becomes positive and statistically significant. The results indicate that listed firms face a trade-off between investing in real estate and productive assets. When expected returns to productive investment are low, growth prospects are weak, or firms experience consecutive losses, they are more inclined to reallocate capital toward real estate to pursue capital gains.

Table 6: Evidence from Listed Firms

	(1)	(2)	(3)
	RE investment	RE investment	RE investment
Ln(SRPs)	0.0055*** (0.0020)	0.0029* (0.0015)	0.0017* (0.0009)
Asset Turnover_dummy	0.0032 (0.0038)		
Ln(SRPs)* Asset Turnover_dummy	-0.0041* (0.0021)		
Tobin Q_dummy		0.0062 (0.0082)	
Ln(SRPs)*Tobin Q_dummy		-0.0035 (0.0022)	
Loss_dummy			0.0519 (0.0333)
Ln(SRPs)*Loss_dummy			-0.0157 (0.0131)
City FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Dep. Var. Mean	0.0334	0.0334	0.0334
R ²	0.9215	0.9216	0.9227
N	1,152	1,152	1,152

Notes: (1) Standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. (2) The dependent variable is firms' real estate investment, measured as the ratio of investment property to total assets. The independent variable is SRPs scale, measured as the logarithm of the announced annual number of newly planned renovation housing units in prefecture c in year t . (3) Asset turnover is defined as the ratio of operating revenue to fixed assets. Tobin's Q is defined as the ratio of market value to total assets. We then construct dummy variables for asset turnover (*Asset Turnover_dummy*) and Tobin's Q (*Tobin Q_dummy*), which equal one if the firm's value exceeds the sample median and zero otherwise. Finally, *Loss_dummy* equals one if a firm reports negative net profits for two consecutive years, and zero otherwise.

5 Housing Booms and Capital Misallocation

5.1 SRPs and TFP Dispersion

In this section, we examine whether the implementation of SRPs leads to local capital misallocation. Specifically, we follow the methodology of Hsieh and Klenow (2009) and measure within-industry capital misallocation using the dispersion in the log of firm-level total factor productivity (TFP).

We adopt three standard dispersion metrics: the interquartile range (P75–P25), the decile range (P90–P10), and the standard deviation of log TFP, all calculated within each prefecture-industry-year cell. Firm-level TFP is estimated using the semiparametric method developed by Olley and Pakes (1996). This method allows us to account for potential simultaneity between input choices and productivity shocks. The regression results are reported in Table 7 columns (1), (3), and (5), where the dependent variable is TFP_dispersion.

Table 7: SRPs and Industry-level TFP Misallocation

	72-25		90-10		Std.	
	(1) TFP_ dispersion	(2) TFP_ dispersion_2020	(3) TFP_ dispersion	(4) TFP_ dispersion_2020	(5) TFP_ dispersion	(6) TFP_ dispersion_2020
Ln(SRPs)	0.0113 (0.0110)		0.0240** (0.0095)		0.0210* (0.0107)	
sum_lnpsl		0.0592** (0.0262)		0.0827*** (0.0231)		0.0679*** (0.0217)
Year FE	Yes	No	Yes	No	Yes	No
Industry FE	Yes	No	Yes	No	Yes	No
Dep. Var. Mean	-1.1098	-1.0229	-0.4977	-0.3953	-1.2734	-1.1593
R ²	0.1356	0.0127	0.1870	0.0189	0.1741	0.0152
N	1,959	611	1,959	611	1,959	611

Notes: (1) Standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. (2) Columns (1), (3), and (5) report panel regressions examining the average effect of SRPs on TFP dispersion. The dependent variable is industry-level TFP dispersion in each year, measured by the interquartile range (P75–P25), the decile range (P90–P10), and the standard deviation of log TFP. The independent variable is the logarithm of the announced annual number of newly planned renovation housing units in prefecture c in year t . Columns (2), (4), and (6) report cross-sectional long-difference regressions. The dependent variable is industry-level TFP dispersion in 2020 measured using the same three indicators, while the independent variable is the logarithm of the total planned SRP construction units in prefecture c over 2016–2020. (3) As a robustness check, we re-estimate firm-level TFP using the Akerberg–Caves–Frazer (ACF) approach (Akerberg et al., 2015) and recompute industry-level TFP dispersion. The results are similar to those reported in this table.

To evaluate the long-run effects of SRPs on capital misallocation, we further explore the relationship between cumulative SRP intensity from 2016 to 2020 and changes in misallocation over the same period in Columns (2), (4) and (6). Our key explanatory variable is defined

as $\text{Ln}(\Delta \text{SRPs}_{2016-2020})$, the logarithm of the total planned SRP construction units in each prefecture over 2016-2020. The dependent variable is the log of firm-level TFP misallocation in 2020, denoted by $\text{TFPdispersion}_{2020}$. Our results show that greater SRP exposure significantly increases the dispersion of firm productivity at the city-industry level. Taking column (3) as an example, a 1% increase in cumulative SRP intensity is associated with a 0.024 increase in within-industry TFP dispersion, which corresponds to 4.82% of its sample mean ($0.024/0.4977$). This magnitude suggests that SRPs contribute meaningfully to the deterioration of capital allocation efficiency across firms.

5.2 SRPs and Underinvestment

If firms shift capital away from physical production toward real estate speculation, their investment levels would fall below the efficient benchmark, leading to underinvestment, which is another measure of inefficiency. To test this conjecture, we construct four firm-level underinvestment measures following the approach of Richardson (2006) and Gan (2007). These indicators capture the gap between a firm's actual net investment and its estimated optimal investment level. Lower values of the indices indicate more severe underinvestment relative to the firm's efficient capital stock. The results presented in Table 8 show that firms located in regions with greater SRP intensity indeed exhibit a significantly greater degree of underinvestment.

Table 8: SRPs and Underinvestment

	(1)	(2)	(3)	(4)
	Underinvest1	Underinvest2	Underinvest3	Underinvest4
Ln(SRPs)	-3.5583* (2.0666)	-3.5366* (2.0526)	-3.5555* (2.0668)	-0.0196*** (0.0060)
City FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Dep. Var. Mean	-0.9176	-0.8804	-0.7883	0.0002
R ²	0.0958	0.0473	0.0458	0.1053
N	139,801	139,801	139,801	111,436

Notes: (1) Standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. (2) The dependent variable is firms' underinvestment level. Columns (1)–(3) follow Richardson (2006) and measure underinvestment as the residual between estimated optimal investment and actual investment. Column (1) uses the lagged firm's sales growth as the proxy for expected optimal investment. Column (2) further includes year and firm fixed effects. Column (3) additionally controls for lagged cash flow, firm age, leverage and firm size when predicting expected investment. Column (4) reports results using the inverse of the overinvestment measure constructed following Gan (2007).

5.3 Further Analysis: TFP and Output

In addition to inducing speculative investment and capital reallocation, housing booms may also stimulate the real economy through a demand channel. Real estate development can generate upstream and downstream spillovers that raise aggregate output, and the liquidity injected through SRPs—primarily in the form of cash-based resettlement—may amplify firm output via local multiplier effects.

To test this possibility, we examine the impact of SRPs on firm-level TFP and output. Our results show that SRPs have significant positive effects on both measures. A 1% increase in SRP intensity is associated with a 0.03% increase in value-added at the firm level. To assess the macroeconomic relevance of this effect, we conduct a simple calculation. Based on our dataset, a 1% increase in SRP activity raises the average value-added per firm by approximately 23.59 RMB (0.03% of the sample mean firm-level value-added of 78,629.99 RMB). Using official statistics from the National Bureau of Statistics of China, which reports an average of approximately 67,289 firms per prefecture over 2016–2020, we aggregate across firms and find that the implied total output gain amounts to approximately 1.58 million RMB per prefecture.⁴

Table 9: SRPs and Firm Output

	(1) Ln(value add)	(2) Ln(TFP)	(3) TFP_growth
Ln(SRPs)	0.0340** (0.0146)	0.0145*** (0.0029)	0.0158*** (0.0033)
City FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Dep. Var. Mean	8.1976	1.4519	-0.0167
R ²	0.8798	0.7498	0.3116
N	62,967	72,225	52,746

Notes: (1) Standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. (2) The dependent variable is the logarithm of firm value added in column (1), firm-level TFP estimated following Olley and Pakes (1996) in column (2), and the logarithmic growth rate of TFP in column (3).

⁴We first recover the original value-added from Ln(value add) in Table 1, yielding a sample mean of 78,629.99 RMB. We then use the average number of enterprise legal entities reported by the National Bureau of Statistics of China (<https://data.stats.gov.cn>) over 2016–2020, which is 19,715,714 nationwide, implying an average of 67,289 firms per prefecture across 293 prefectures. The aggregate effect is computed as $67,289 \times 23.59 \approx 1.58$ million RMB.

6 Conclusion

This paper addresses an underexplored question in the literature on housing booms and firm behavior: whether and how policy-driven housing price surges reshape capital allocation within firms, beyond the well-documented credit-based channels (collateral effects and credit crowding-out). Taking China’s Shantytown Renovation Programs (SRPs), particularly the post-2015 shift to cash-based resettlement, as a policy shock, we focus on identifying the within-firm capital allocation channel and evaluating its impacts on productive investment, real estate speculation, and aggregate resource allocation efficiency. This research fills the gap in understanding how housing policies affect micro-level firm decisions and productive efficiency through internal capital reallocation.

Our empirical results yield three key findings. First, SRP-induced housing booms significantly reduce firms’ productive asset investment: an annual increase of 1% planned SRP units leads to a 0.27% decline in local firms’ productive investment, according to our preferred instrumental variable regressions. Second, this decline is not explained by collateral constraints or reduced credit supply; instead, firms actively reallocate capital toward real estate investment, especially those with lower returns on productive assets, verifying the within-firm capital allocation channel as an independent mechanism. Third, such reallocation causes resource misallocation: SRPs increase prefecture-industry TFP dispersion and raise firm underinvestment, though they generate modest output gains through demand-side effects.

This study has limitations that point to avenues for future research. First, data constraints restrict our analysis: we lack more granular SRP data (e.g., the exact share of cash-based resettlement, compensation amounts per household) and detailed firm-level real estate holdings (e.g., property type, location, and investment motives), which could deepen insights into heterogeneous effects. Second, future research could extend the focus beyond firms to explore SRPs’ impacts on household finance and consumption—for instance, how cash-based resettlement affects households’ asset allocation, consumption structure, and savings behavior, as well as the spillover effects between household and firm decisions. Such extensions would provide a more comprehensive understanding of the economic consequences of large-scale housing renovation programs.

Declaration of generative AI technologies in the writing process

During the preparation of this work, the authors used GPT 4o developed by OpenAI in order to improve readability and language. After using this tool/service, the authors reviewed and edited the content as needed and assume full responsibility for the content of the publication.

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